

Exercice (1)

Q1 $\vec{A} = -\frac{4}{3}\pi R^3 \rho_a \vec{g}$

Q2 $\left| \frac{\vec{A}}{\vec{P}} \right| = \frac{\rho_a}{\rho_e} = 10^{-3}$

donc $\|\vec{A}\| \ll \|\vec{P}\|$

On peut négliger la poussée d'Archimède

Q3 système: goutte / réf: terre/gelée

$$\frac{4}{3}\pi R^3 \rho_e \frac{d\vec{v}}{dt} = \frac{4}{3}\pi R^3 \rho_e \vec{g} - 6\pi \eta R \vec{v}$$

$\downarrow \vec{e}_z$

$$\frac{dv}{dt} + \frac{9}{2} \frac{\eta}{R^2 \rho_e} v = g$$

Q4 $\tau = \frac{2R^2 \rho_e}{9 \eta}$

A.N. $\tau =$

Q5

$$v(t) = \underbrace{\lambda \exp\left(\frac{t}{\tau}\right)}_{\text{sh}} + \underbrace{g\tau}_{\text{sp}} \quad (\text{EO ordre 1})$$

or $v(t=0) = 0 \Rightarrow \lambda = -g\tau$

d'où $v(t) = g\tau \left(1 - \exp(-t/\tau)\right)$

Q6

$$v_e = g\tau$$

A.N. $v_e =$

Q7

$$\frac{dv}{dt} = \alpha S(t)$$

$$\frac{4}{3}\pi \frac{dn^3}{dt} = \alpha 4\pi r^2$$

$$\frac{dn^3}{r^2} = 3\alpha dt$$

$$\frac{3n^2 dr}{r^2} = 3\alpha dt$$

$$dr = \alpha dt$$

$$r(t) = \alpha t + r_0$$

r_0 rayon initial.

$$\alpha = k$$

Q8 $m = \frac{4}{3} \pi r^3(t) \rho_e$

d'où $\frac{dm}{dt} = \frac{4}{3} \pi \rho_e \frac{dr^3}{dt}$

$\frac{dm}{dt} = \frac{4}{3} \pi \rho_e \frac{d}{dt} (r_0 + kt)^3$

$\frac{dm}{dt} = \frac{4}{3} \pi \rho_e \cdot 3k (r_0 + kt)^2$

Q9 Remarquez d'abord que la Q2.

~~$m \frac{dv}{dt} + \frac{dm}{m dt} v = m g - \frac{6\pi b_a r(t)}{m} v$~~

$\frac{dv}{dt} + \left[\frac{1}{m} \frac{dm}{dt} + \frac{6\pi b_a r(t)}{m} \right] v = g$

$\frac{1}{m} \frac{dm}{dt} = \frac{\frac{4}{3} \pi \rho_e \cdot 3k r(t)^2}{\frac{4}{3} \pi \rho_e r(t)^3} = \frac{3k}{r_0 + kt}$

$A = 3k$

~~$\frac{36\pi b_a r(t)}{2\pi r(t)^2 \rho_e} = \frac{3 b_a}{(r_0 + kt)^2}$~~

avec $B = \frac{3 b_a}{2 \rho_e}$

$\frac{dv}{dt} + \left(\frac{A}{r(t)} + \frac{B}{r(t)^2} \right) v = g$

Q10 $\frac{dv}{dt} + \frac{A}{r(t)} v(t) = g$ (*)

EH. $\frac{dv_h}{dt} = -A \frac{v_h}{r(t)} dt$: séparation des variables

$\ln(v_h) = \left(-\frac{A}{k} \right) \ln(r_0 + kt) + C$
 expl. et $C_2 = \exp(C)$.

$v_h = C_2 (r_0 + kt)^{-A/k}$ or $A = 3k$

$v_h = C_2 (r_0 + kt)^{-3}$

solut° particulier $v_p = \lambda t + \mu$

dans (*) $\lambda + \frac{A}{r_0 + kt} (\lambda t + \mu) = g$

$\lambda(r_0 + kt) + A(\lambda t + \mu) = g(r_0 + kt)$

variable x et $d'x$

$$\begin{cases} \lambda r_0 + \mu = g r_0 \\ \lambda k + \mu = g k \end{cases} \Rightarrow \begin{cases} \mu = \frac{r_0}{k} (g - \lambda) \\ \lambda = \frac{g k}{k + r_0} \end{cases}$$

$$\begin{cases} \lambda = \frac{g k}{3k + k} = \frac{g}{4} \\ \mu = \frac{r_0}{3k} \frac{g}{4} = \frac{g r_0}{4k} \end{cases}$$

au final

$$v(t) = v_h + v_p$$

$$v(t) = \frac{C_2}{(r_0 + kt)^3} + \frac{g}{4} \left(t + \frac{r_0}{k} \right)$$

$$v(t=0) = 0 \Rightarrow C_2 = -\frac{g}{4} \frac{r_0^4}{k}$$

$$v(t) = \frac{g}{4} \left(t + \frac{r_0}{k} - \frac{r_0}{k} \frac{1}{(r_0 + kt)^3} \right)$$

Q11 Il s'agit de la tension superficielle.